

Drone Build Overview:

Introduction:

In this report, I will be describing a custom-built drone designed for aerial flight. The drone utilizes a combination of key components such as a 2200mAh battery, 30A ESCs, 2200KV BLDC motors, the KK2.1.5 flight controller, and a FlySky transmitter and receiver system. Each of these components plays a vital role in the overall performance and functionality of the drone.

1. Power System:

- **Battery (2200mAh):**

The 2200mAh battery is a key component that provides the necessary power for the drone to operate. It typically comes in a 3S or 4S configuration, with 3S (11.1V) being more common for mid-range drones. The capacity of 2200mAh indicates the battery's energy storage capability, which will affect flight time, weight,

and power output. Depending on the motor and propeller setup, the flight time may range from 6 to 15 minutes, though this varies based on conditions such as payload weight and flight style.

- **ESC (30A):**

The ESCs (Electronic Speed Controllers) are responsible for regulating the power sent to the drone's motors. The 30A rating means that each ESC can handle up to 30 amps of current, which is suitable for your 2200KV BLDC motors. The ESCs convert the power from the battery into a form that the motors can use, allowing for control over the drone's speed and responsiveness. Having the right rating ensures that your motors receive the appropriate amount of power without overloading the ESCs.

2. Motors:-

- **2200KV BLDC Motors:**

The motors chosen for this drone are 2200KV brushless DC (BLDC) motors. The KV rating (2200KV) indicates how fast the motor can spin per volt applied to it—essentially, the RPM per volt. A higher KV rating

like this suggests that the motor is designed for higher-speed applications, which will be useful for quicker response times, higher agility, and more thrust for lifting off. The combination of 2200KV motors with a 30A ESC allows the drone to achieve powerful thrust-to-weight ratios, ideal for both recreational and racing purposes.

3. Flight Control System:-

- **KK2.1.5 Flight Controller:**

The KK2.1.5 flight controller is a popular choice for hobbyists looking for a simple, user-friendly setup. This flight controller is equipped with an integrated LCD screen for easy setup and tuning. The KK2.1.5 provides basic stabilization and control algorithms that allow for stable flight. It supports various flight modes, including manual, self-leveling, and altitude hold, and is compatible with many types of drones, including multirotors (quads, hexacopters, etc.). The controller communicates with the ESCs to adjust motor speeds and maintain stable flight based on the pilot's inputs and sensor data.

4. Communication System:

- **FlySky Transmitter and Receiver:**

The FlySky transmitter and receiver system enables wireless communication between the pilot and the drone. The transmitter sends control signals (throttle, yaw, pitch, and roll) to the receiver, which then relays the commands to the flight controller. FlySky systems are known for their reliability, ease of use, and relatively low cost, making them an excellent choice for hobbyists. The range of the system depends on the specific model, but many FlySky systems offer a solid range (up to several hundred meters) for most recreational drone flying.

8 5. Additional Considerations:

- **Propellers:**

While you haven't mentioned the propellers specifically, it's important to note that the size and pitch of the propellers will influence the drone's flight characteristics. Typically, drone propellers are sized based on the motor and battery setup. For your 2200KV motors, a typical choice might be 5-6 inch

propellers for good balance between thrust and efficiency.

- **Frame:**

The frame of the drone, not specified here, should be lightweight yet sturdy enough to support all the components. Common materials for drone frames include carbon fiber and plastic. The frame size and design will affect the drone's stability and maneuverability, and it should be able to accommodate the size of the motors and battery.

6. Performance:

With this configuration, the drone should offer excellent performance for most recreational flying, freestyle, or racing applications. The 2200KV motors and 30A ESCs provide a high thrust-to-weight ratio, enabling fast accelerations and responsive controls. The KK2.1.5 flight controller offers sufficient stability and tuning flexibility for a wide range of flying conditions, from smooth cruising to more aggressive flying maneuvers.

7. Conclusion:

In conclusion, this drone setup offers a solid balance of power, control, and ease of use. The 2200mAh battery provides ample power for extended flight times, while the 2200KV BLDC motors and 30A ESCs ensure fast and responsive flight. The KK2.1.5 flight controller offers good flight stability and customization, and the FlySky transmitter and receiver ensure reliable control over the drone. Together, these components create a well-rounded drone capable of both fun recreational flying and more performance-focused applications.

Speech Example:

"Good [morning/afternoon], everyone. Today, I'm going to present a custom-built drone that I've assembled, showcasing the various components that contribute to its overall performance."

At the heart of this drone is a 2200mAh battery, providing the necessary power to keep the motors running. Paired with this is a set of 30A ESCs, which control the speed and power output to the motors, ensuring smooth and responsive flight. These 2200KV brushless DC motors are designed for high-speed operation, delivering both agility and thrust.

The flight controller I've chosen for this build is the KK2.1.5, a reliable and straightforward system that stabilizes the drone and helps maintain control in various flight conditions. The KK2.1.5 is versatile, supporting multiple flight modes, making it ideal for both beginners and more advanced drone pilots.

To communicate with the drone, I've used a FlySky transmitter and receiver system, which ensures a reliable connection and control over the drone during flight.

With these components working together, this drone is capable of delivering a solid performance, whether you're just starting out or looking to perform more advanced maneuvers.

Thank you for your attention, and I hope this gives you a better understanding of the components that make up my drone build."